

FINANCIAL LITERACY

PATHWAYS TO ARTS & HUMANITIES SUMMER SCHOLARS PROGRAM

Instructor: Tudor Schlanger

Summer 2026

WHAT IS FINANCIAL LITERACY?

Financial Literacy

The ability to understand and use everyday money skills, e.g. **saving**, **borrowing**, and **investing**.

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The ability to understand and use everyday money skills, e.g. **saving**, **borrowing**, and **investing**.

Why does it matter?

- Everyone deals with money.
- Making good decisions builds wealth.
- Mistakes are expensive!



QUIZ: QUESTION 1 OF 4

Question

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have in the account if you left the money to grow?

- A. More than \$102
- B. Exactly \$102
- C. Less than \$102
- D. Do not know

QUIZ: QUESTION 2 OF 4

Question

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- A. More than today
- B. Exactly the same
- C. Less than today
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QUIZ: QUESTION 3 OF 4

Question

If interest rates rise, what should happen to bond prices?

- A. They should rise
- B. They should fall
- C. They should stay the same
- D. There is no relationship between bond prices and interest rates
- E. Do not know

QUIZ: QUESTION 4 OF 4

Question

Please tell me whether this statement is true or false: “Buying a single company’s stock usually provides a safer return than a stock mutual fund.”

- A. True
- B. False
- C. Do not know

OUTLINE

1. Saving & Compound Interest

2. Inflation

3. Bonds & Interest Rates

4. Stocks, ETFs & Diversification

5. Recap

WHAT IS SAVING?

Saving

Setting aside part of your income today so you can use it in the future.

Why do we need to save?

- Big future purchases (college, car, house)
- Retirement (you won't work forever!)
- Emergencies (unexpected expenses)



Photo: Pexels (free license)

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How do we **save**?



Photo: Pexels (free license)

Answer

By putting money into **assets**.

TYPES OF ASSETS

Financial Assets

- Savings accounts
- Bonds
- Stocks

Non-Financial Assets

- House
- Car
- Land

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How to tell them apart:

- **Tangibility:** non-financial assets are physical, you can touch a house, but not a stock
- **Liquidity:** financial assets are easy to sell quickly (click a button); selling a house takes months

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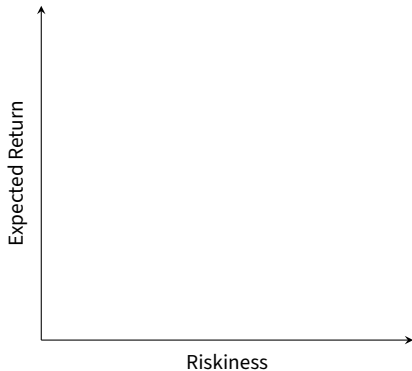
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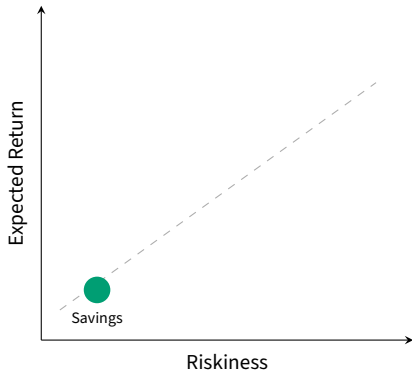
How are these financial assets different?

FINANCIAL ASSETS



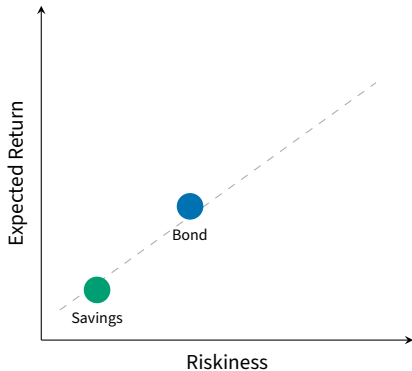
- **Expected return:** how much you *typically* expect your money to grow. Put in \$100, get back \$105 in a year? That's a 5% return.
- **Riskiness:** how *uncertain* that return is. A safe bet pays about what you expect; a risky one might soar...or lose money.

FINANCIAL ASSETS



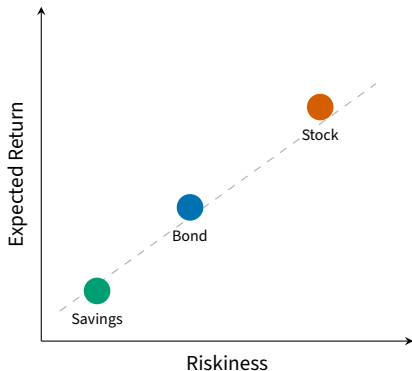
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FINANCIAL ASSETS



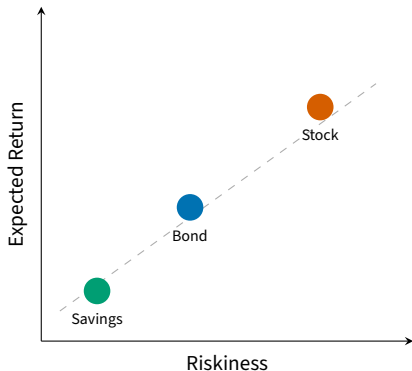
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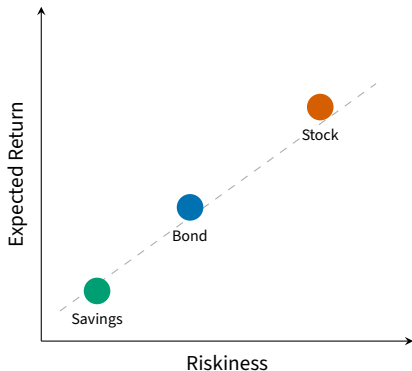
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Why do we care about higher expected returns?

COMPOUND RETURNS

Year 1:

$$A_1 = P \times (1 + r)$$

You earn a return on your starting amount.

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The second year's return is applied to the **whole** Year 1 balance — not just the original P .

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In general:

$$A_t = P \times (1 + r)^t$$

You earn returns on your returns. Each year, the growth accelerates.
This is **compounding**.

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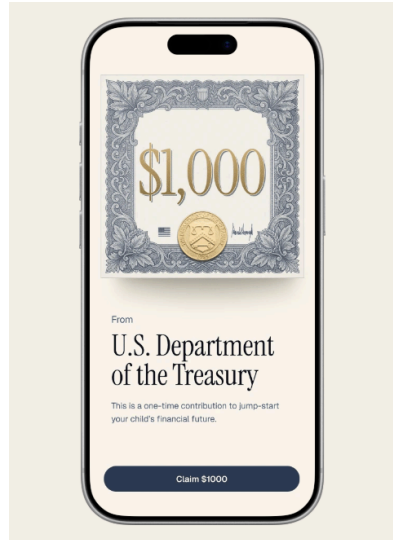
So if we want to save a lot, we should start early!

POLICY: TRUMP ACCOUNTS

Trump Accounts

The government deposits **\$1,000** into an investment account for every newborn. The child can access the money at **age 18**.

- These accounts invest money in the stock market.



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Question

Hypothetically, these accounts give a rate of return of 10%. How much would \$1,000 grow to in 18 years?



COMPOUND RETURNS

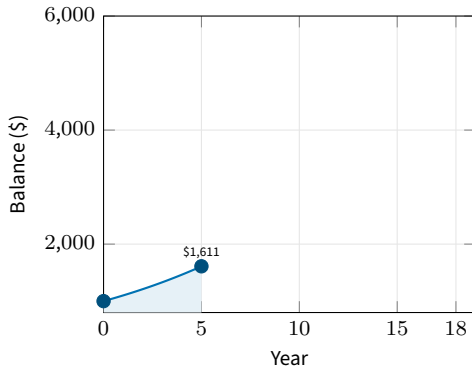
$$A_t = P \times (1 + r)^t$$

P = starting amount r = rate of return t = years

Example: \$1,000 at 10% per year.

After 5 years:

$$A = 1,000 \times 1.10^5 = \text{\textcolor{teal}{\$1,611}}$$



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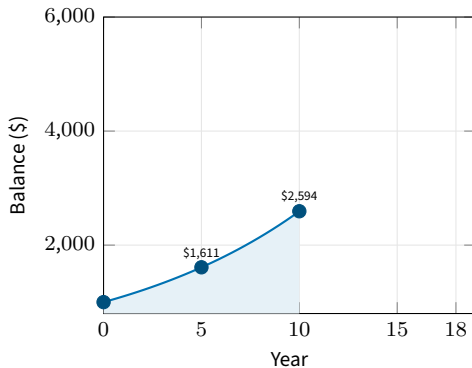
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After 10 years:

$$A = 1,000 \times 1.10^{10} = \text{\textcolor{teal}{\$2,594}}$$



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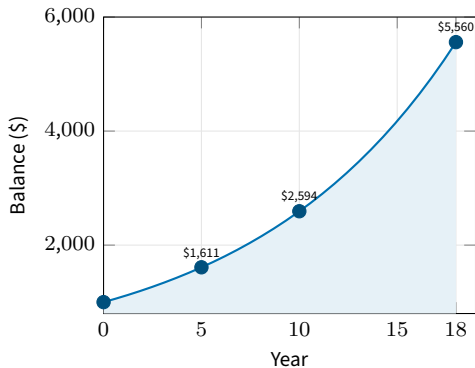
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After 10 years: $A = \$2,594$

After 18 years:

$$A = 1,000 \times 1.10^{18} = \text{\textcolor{blue}{\$5,560}}$$



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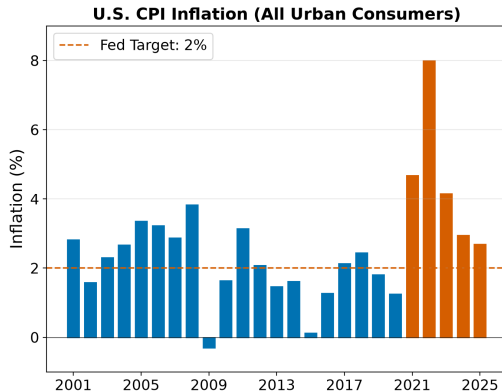
WHAT IS INFLATION?

Inflation

A sustained increase in the general price level of goods and services. When prices rise, each dollar buys **less**.

Example:

- A coffee costs \$3 today.
- If inflation is 5%, next year it costs \$3.15.
- Your dollar lost **purchasing power**.



Source: FRED, CPI-U (All Urban Consumers)

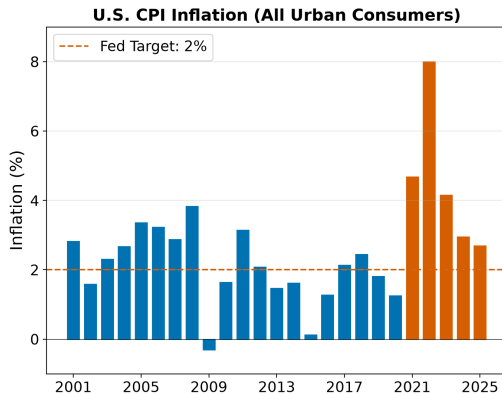
MEASURING INFLATION: THE CPI

Consumer Price Index (CPI)

The CPI tracks the cost of a fixed “basket” of goods and services (food, housing, gas, health-care...) that a typical household buys.

- The **Bureau of Labor Statistics (BLS)** measures it monthly
- You can explore the data yourself at **bls.gov** and **fred.stlouisfed.org**

If the CPI was 100 last year and is 103 this year, inflation was 3%.

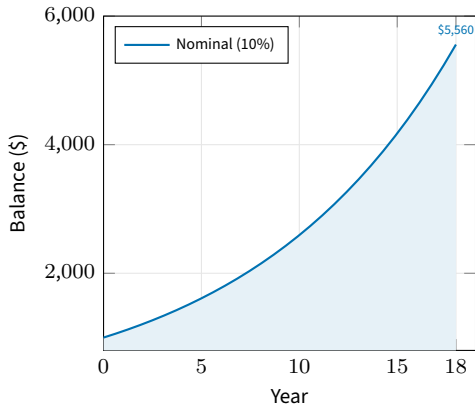


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REAL VS. NOMINAL RETURNS

Real return $\approx$ Nominal return – Inflation

Example: Your investment returns 10% (nominal).
Inflation is 5%.



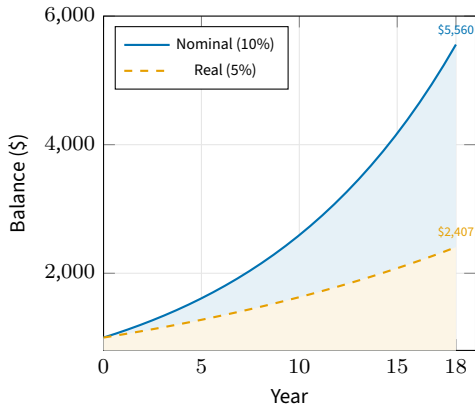
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Example: Your investment returns 10% (nominal).
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Real return $\approx 10\% - 5\% = 5\%$

Inflation eats into your returns. What matters is the **real return**, not the nominal one.



ANSWER TO Q1

Question

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have?

- A. More than \$102
- B. Exactly \$102
- C. Less than \$102
- D. Do not know

ANSWER TO Q1

Question

Suppose you had \$100 in a savings account and the interest rate was 2% per year. After 5 years, how much do you think you would have?

- A. More than \$102
- B. Exactly \$102
- C. Less than \$102
- D. Do not know

Answer

A. More than \$102. With compound returns, you earn returns on your returns. After 5 years at 2%:

$$100 \times 1.02^5 = \$110.41$$

Even in just 5 years, compounding adds more than the \$2 from the first year alone.

ANSWER TO Q2

Question

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- A. More than today
- B. Exactly the same
- C. Less than today
- D. Do not know

ANSWER TO Q2

Question

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, how much would you be able to buy with the money in this account?

- A. More than today
- B. Exactly the same
- C. Less than today
- D. Do not know

Answer

C. Less than today. With 1% interest and 2% inflation, your real return is approximately -1% . Your account balance grows to \$101, but the things you want to buy now cost \$102. You can buy **less** than before.

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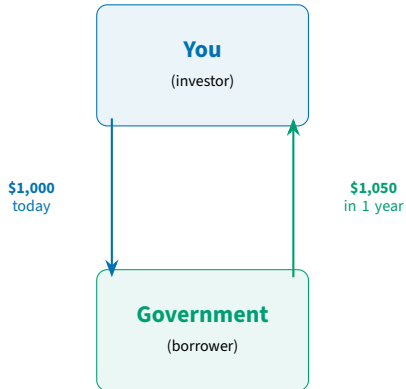
WHAT IS A BOND?

Bond

A bond is a **loan you make** to the government (or a company). They promise to pay you back with interest after a fixed period.

Think of it as an IOU:

- You give the government \$1,000 today
- In 1 year, they pay you back \$1,050 (that's 5% interest)
- Your return: \$50



Bonds are considered safe because the government almost never fails to pay you back.

BOND PRICING

$$P = \frac{\text{Payout}}{1 + r}$$

P = price today Payout = paid in 1 year r = interest rate

Example: A bond that pays \$1,050 in one year.

- If $r = 5\%$: $P = \frac{1,050}{1.05} = \$1,000$

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- If r rises to 7%: $P = \frac{1,050}{1.07} \approx \981

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- If r rises to 7%: $P = \frac{1,050}{1.07} \approx \981
- If r drops to 3%: $P = \frac{1,050}{1.03} \approx \$1,019$

Bond prices and interest rates move in **opposite directions**. This is one of the most important relationships in finance.

ANSWER TO Q3

Question

If interest rates rise, what should happen to bond prices?

- A. They should rise
- B. They should fall
- C. They should stay the same
- D. There is no relationship between bond prices and interest rates
- E. Do not know

ANSWER TO Q3

Question

If interest rates rise, what should happen to bond prices?

- A. They should rise
- B. They should fall
- C. They should stay the same
- D. There is no relationship between bond prices and interest rates
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Answer

B. They should fall.

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WHAT IS THE STOCK MARKET?

Stock Market

A marketplace where people buy and sell **shares** — small pieces of ownership in companies like Amazon or Google

When you buy a share of Amazon, you literally **own a tiny piece** of the company.

- If Amazon does well → your share becomes more valuable
- If Amazon does poorly → your share loses value



Photo: Pexels (free license)

SINGLE STOCKS: THE UPSIDE

Amazon went public in 1997 for ~\$0.1 per share.

By 2025: ~\$200 per share.

That's a factor of $\times 2000$!

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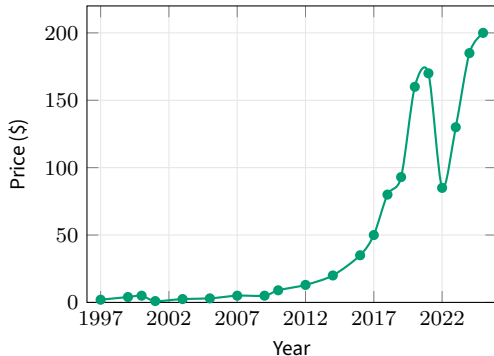
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Question

What is the stock price of a company that goes bankrupt?



Approximate split-adjusted prices

SINGLE STOCKS: THE DOWNSIDE

Enron was a Houston-based energy company that traded electricity, natural gas, and other commodities.

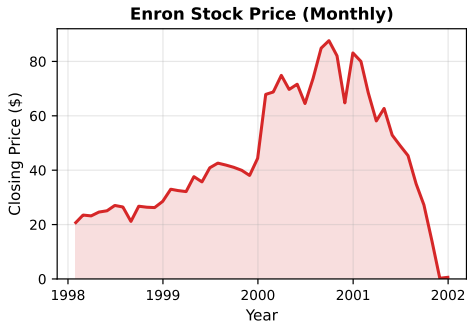
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- Over 20,000 employees
- Named “America’s Most Innovative Company” by *Fortune* — six years in a row

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In 2001, it was revealed that executives had **hidden billions in debt** through fake partnerships. The stock went from \$90 to **zero**.



A BETTER WAY: FUNDS

Fund

A basket of many stocks bundled together. When you buy one share of a Fund, you own a tiny piece of **hundreds** (or thousands) of companies.

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- **Example:** the S&P 500 ETF (SPY) holds shares of the 500 largest U.S. companies



Large public companies

THE POWER OF DIVERSIFICATION

Single Stock

Company fails
→ You lose **everything**

High risk

vs.

ETF (500 stocks)

One company fails
→ Barely noticeable

Lower risk

Diversification means spreading your money across many investments. If one fails, the others cushion the blow.

Instead of gambling on one company, you are now betting on the whole economy. This is still a risk!

ANSWER TO Q4

Question

Please tell me whether this statement is true or false: “Buying a single company’s stock usually provides a safer return than a stock mutual fund.”

- A. True
- B. False
- C. Do not know

ANSWER TO Q4

Question

Please tell me whether this statement is true or false: “Buying a single company’s stock usually provides a safer return than a stock mutual fund.”

- A. True
- B. False
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Answer

B. False. A single stock is **riskier** than a mutual fund.

A mutual fund holds many stocks, so if one company fails, the impact on your portfolio is small. This is **diversification** — one of the most important concepts in investing.

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Let's see which team won!

TEAM RESULTS: THE QUIZ

How did each team score on the four questions?

Place	Team	Correct (out of 4)
4th	Team D	1

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Answer

Congratulations, Team A!

FINANCIAL LITERACY IN THE U.S.

Table 2

Financial Literacy in the US Population: Big Three Questions

	<i>Correct</i>	<i>Incorrect</i>	<i>Do not know/ refuse</i>
Interest	80.6%	16.4%	3.0%
Inflation	75.5%	20.7%	3.8%
Risk	60.7%	17.1%	22.2%
All Big Three correct	43.3%		

Source: Authors' tabulations, 2019 Survey of Consumer Finance.

Note: All data weighted using sampling weights.

WHAT CAN YOU DO?

1. **Start saving early** — Compound interest rewards patience.
2. **Watch inflation** — Make sure your returns beat it.
3. **Understand bonds** — They're safe and good for stability.
4. **Diversify with Mutual Funds or ETFs** — Don't bet everything on one company.

Thank you!



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